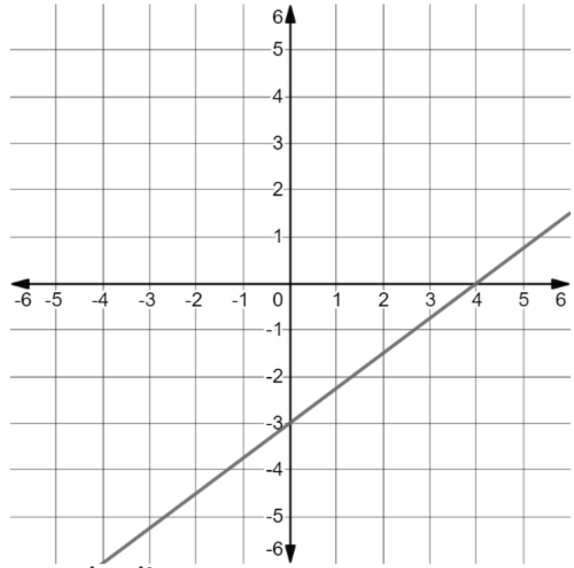
Algebra 1 Name____________________
Unit 13 Date____________________
Lesson H1 Practice Period__________________

The graph of the function $G(x)$ is shown.



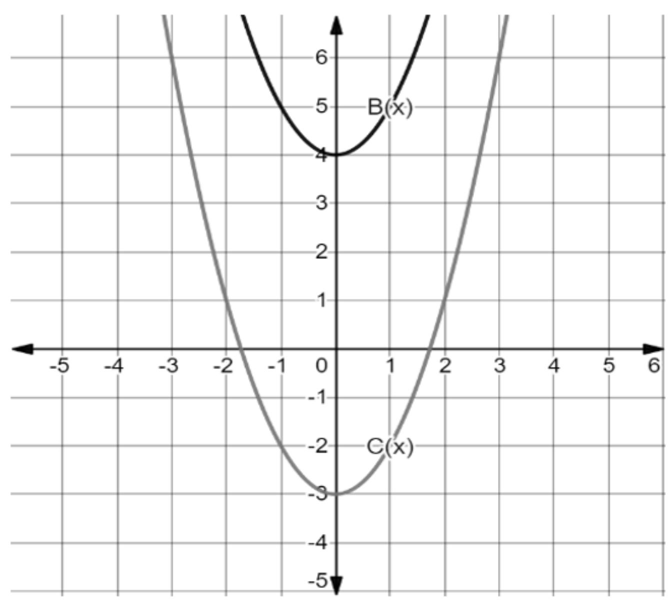
1. Write the function, $G(x)$, that represents the line.

2. Draw a line that is parallel to $G(x)$ with a y-intercept of 3. Label this line $H(x)$.

3. Describe how $G(x)$ can be shifted to coincide with $H(x)$.

4. Write the function $H(X)$ in terms of $G(x)$.

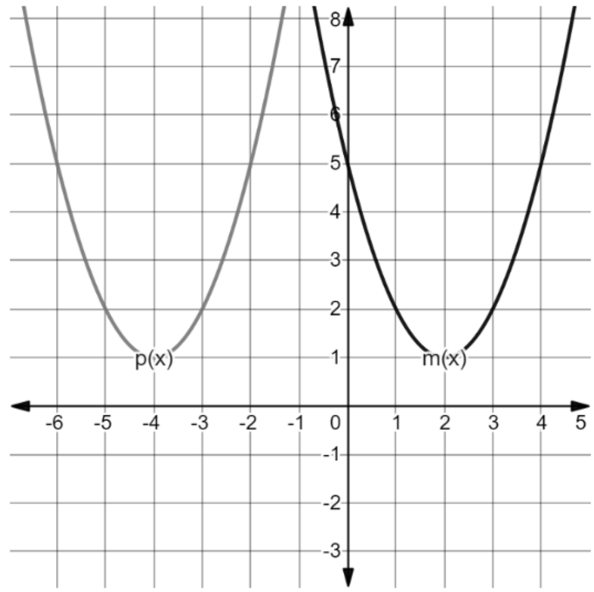
5. The graphs of $B(x)$ and $C(x)$ are shown on the graph.



Write the transformation by completing the equation.

$$C(x) = B(x) + \_\_\_\_$$

The functions $m(x)$ and $p(x)$ are shown on the graph.

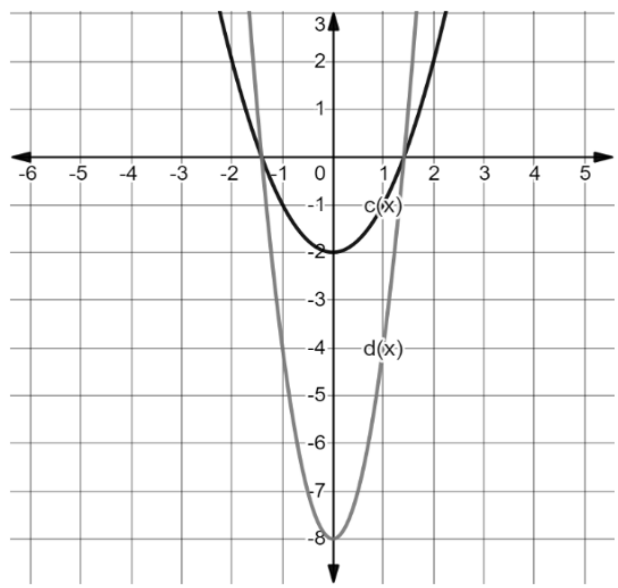


6. Write $m(x)$ in vertex form.

7. Write $p(x)$ in vertex form.

8. Describe how $m(x)$ was transformed to create $p(x)$.

9. The functions $c(x)$ and $d(x)$ are shown on the graph, where $d(x)$ is a transformation of $c(x)$. What is the value of k such that $d(x) = kc(x)$?



10. The table of values for $d(x)$ and $f(x)$ are shown, where $f(x)$ is a transformation of $d(x)$. Write a description of the transformation.

$y = d(x)$		$y = f(x)$	
x	y	x	y
-2	16	-2	20
-1	4	-1	8
0	0	0	4
1	4	1	8
2	16	2	20